1 Measurement setup

1.1 Location

Ernst-Abbe-Zentrum
Ehrenbergstraße 29
98693 Ilmenau, Germany

1.1.1 Setup at EAZ site

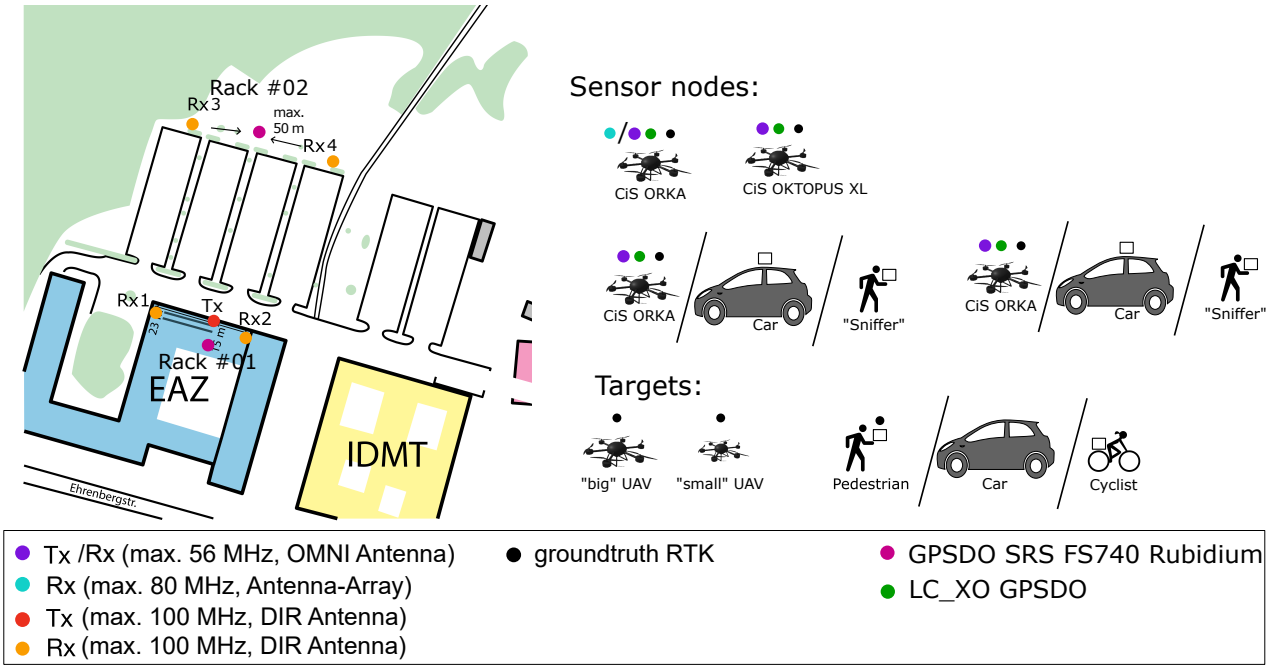


Figure 1.1: Illustration of antenna placement for stationary sensor nodes at rooftop of the EAZ building and mobile sensor nodes / targets

2 Hardware

Mobile SISO sensor node

- 4x UAV sensor node at CiS ORKA
- 1x portable sensor node for pedestrian/bike with helmet
- 2x portable sensor node for "Car" *mag-mount antennas

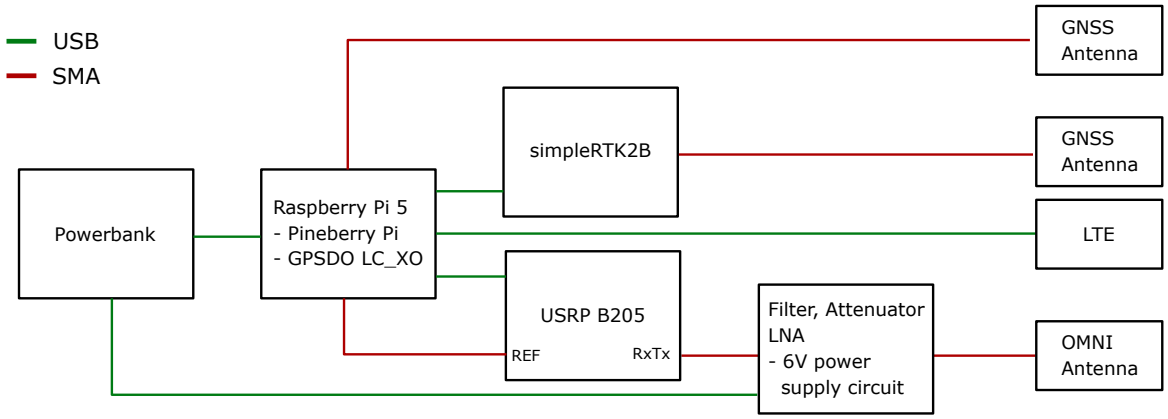


Figure 2.1: Schematic illustration of connections at the mobile SISO node with omni-directional antenna.

Component	Number	Weight (g)	Dimensions (mm)	Power supply
3D printed Case	1	880	170x235x145	
Powerbank[1]	1 (+1)	400		20.000mAh @ 5V
Host PC (Raspberry Pi 5 [2])	1	125		
incl. Fan	1			
Pineberry Pi (2TB SSD) [3]	1 (+1)			
ca. 2h with 56 MS/s Rx				
GPSDO LCXO switch board	1			
Cable USB 3.0 A - C	1	10	300	
Cable SMA-SMA (10 MHz/1PPS)	1	20	300	
GNSS Antenna[4] (L1/L2/L5 helical)	1	20		
GNSS Antenna mount	1	5		

[1] https://www.berrybase.de/quickcharge-powerbank-mit-qc3.0-power-delivery-und-usb-c-20.000-mah-schwarz
[2] https://www.raspberrypi.com/products/raspberry-pi-5/
[3] https://pineberrypi.com/products/hatdrive-bottom-2230-2242-2280-for-rpi5
[4] https://www.ardusimple.com/product/helical-antenna/

Cable SMA-SMA (GNSS)	1	50	1m	
* mag-mount GNSS Antenna *	1			
LTE USB-Stick [5]	1	35		
SIM card [6]	1			
Cable USB A - A socket	1	30	1m	
simpleRTK2B[7] (u-blox F9P)	1	25		
GNSS Antenna[8] (L1/L2/L5 helical)	1	20		
GNSS Antenna mount	1	5		
Cable SMA-SMA (GNSS)	1	50	1m	
* mag-mount GNSS Antenna *	1			
Cable USB A - Micro	1	10	300	
USRP B205mini[9]	1	110		<900 mA @ 5V
Cable USB 3.0 A - Micro B	1	15	260	
LNA 0.5-8 GHz (ZX60-83LN-S+[10])	1	60		<94 mA @ 6V
incl. 6V power supply - USB	1			
Adapter SMA - SMA	1	5		
Tx:				
Highpass [11](3.9-9.8 GHz)	1	10		
Rx:				
6 dB attenuator SMA	1	5		
Highpass [12](2.9-8.7 GHz)	1	10		
Cable SMA-SMA (HF)	1	20	300	
Antenna-mount * mag-mount	1	20		
Omni-Antenna (by Dr.Ing. Blau)	1	40		

Table 2.1: Hardware components mobile SISO sensor nodes

Mobile 4-channel sensor node

- 1x UAV sensor node at CiS ORKA

[5] https://consumer.huawei.com/en/mobile-broadband/e3372/
[6] https://geschaeftskunden.telekom.de/startseite/mobilfunk/462274/data-comfort-xs-business.html
[7] https://www.ardusimple.com/simplertk2b/
[8] https://www.ardusimple.com/product/helical-antenna/
[9] https://www.ettus.com/all-products/usrp-b205mini-i/
[10] https://www.minicircuits.com/pdfs/ZX60-83LN-S+.pdf
[11] https://www.minicircuits.com/pdfs/VHF-3500+.pdf
[12] https://www.minicircuits.com/pdfs/VHF-2700A+.pdf

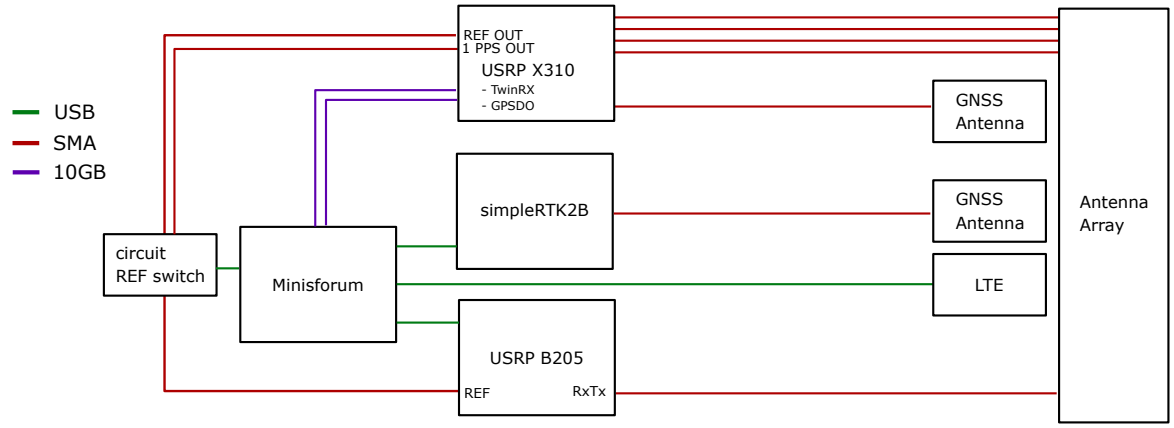


Figure 2.2: Schematic illustration of connections at the mobile node with antenna array.

Component	Number	Weight (g)	Dimensions (mm)	Power supply
3D printed Case	1	660		
LiPo battery	1	815		
circuit - power supply	1			
Host PC Minisforum MS01 (8TB SSD, without housing -> housing +550g weight)	1	850		19V 2.4A
LTE USB-Stick [13]	1	35		
SIM card [14]	1			
Cable USB A - A socket	1	30	1m	
simpleRTK2B[15] (u-blox F9P)	1	25		
GNSS Antenna[16] (L1/L2/L5 helical)	1	20		
GNSS Antenna mount	1	5		
Cable SMA-SMA (GNSS)	1	50	1m	
Cable USB A - Micro	1	15	300	
USRP X310 w TwinRX [17] (without cover -> cover +260g weight) w GPSDO LCXO	1	2000		12V 3.5A
GNSS Antenna[18] (L1/L2/L5 helical)	1	20		
GNSS Antenna mount	1	5		

[13] https://consumer.huawei.com/en/mobile-broadband/e3372/
[14] https://geschaeftskunden.telekom.de/startseite/mobilfunk/462274/data-comfort-xs-business.html
[15] https://www.ardusimple.com/simplertk2b/
[16] https://www.ardusimple.com/product/helical-antenna/
[17] https://www.ni.com/docs/de-DE/bundle/usrp-2945-specs/page/specs.html
[18] https://www.ardusimple.com/product/helical-antenna/

Cable SMA-SMA (GNSS)	1	50	1m
Cable SMA-SMA (10 MHz/1PPS)	3	10	300
switch circuit to REF USRP B205	1	30	
Cable 10GB	2	60	0.5m
Cable SMA-SMA (HF)	4	20	300
USRP B205mini[19] (without housing -> housing +40g weight)	1	70	
Cable USB 3.0 A - Micro B	1	15	260
Cable SMA-SMA (HF)	1	25	400
Antenna-Array (Dr.Ing. Blau) antenna mount + 5 SMA-SMA Winkel	1	105	

Table 2.2: Hardware components mobile 4-channel sensor nodes

Mobile Ground Truth (e.g. for targets)

- 2x UAV (DJI Matrice, VTOL)
- 2x portable for pedestrian, car, cyclist, ... *mag-mount GNSS Antenna at "Helm" instead of helical

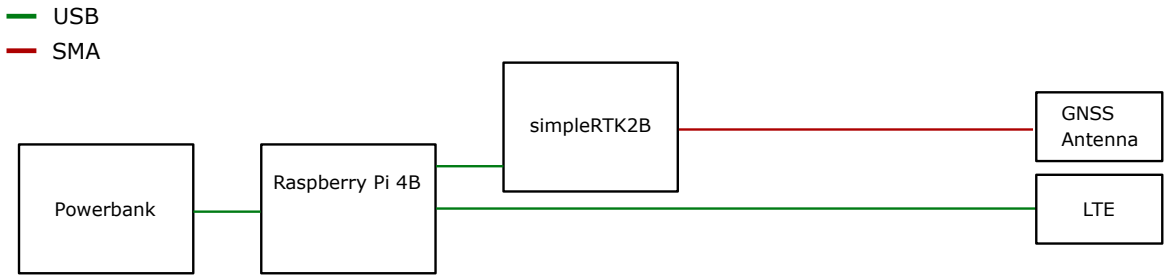


Figure 2.3: Schematic illustration of connections at the target node with RTK.

Component	Number	Weight	Dimensions (mm)	Power supply
3D printed Case	1	275		
Powerbank[20]	1 (+1)	230		10.000mAh @ 5V
Host PC (Raspberry Pi 4B [21])	1	155		
Cable USB 3.0 A - C	1	20		
LTE USB-Stick [22]	1	35		
SIM Karte[23]	1			

[19] https://www.ettus.com/all-products/usrp-b205mini-i/
[20] https://www.amazon.de/Xtorm-Essential-Powerbank-XE1101-geeignet/dp/B0C1WL5YNP
[21] https://www.berrybase.de/raspberry-pi-4-computer-modell-b-8gb-ram
[22] https://consumer.huawei.com/en/mobile-broadband/e3372/
[23] https://geschaeftskunden.telekom.de/startseite/mobilfunk/462274/data-comfort-xs-business.html

Cable USB A - A socket	1	30	1m
simpleRTK2B[24] (u-blox F9P)	1	25	
GNSS Antenna[25] (L1/L2/L5 helical)	1	20	
GNSS Antenna mount	1	5	
Cable SMA-SMA (GNSS)	1	50	1m
* mag-mount GNSS Antenna and "Helm" *	1		
Cable USB A - Micro	1	15	300

Table 2.3: Hardware components mobile Ground Truth

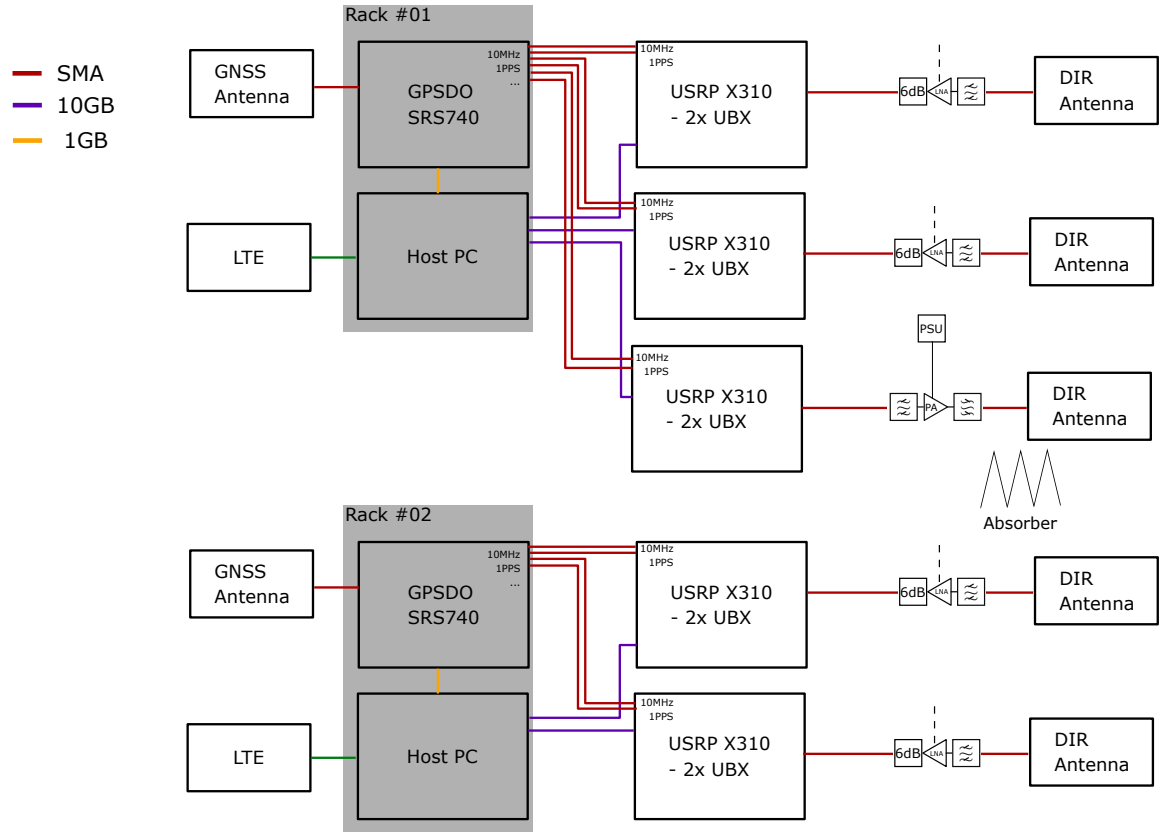


Figure 2.4: Schematic illustration of connections at the stationary nodes at the EAZ rooftop and parking lot.

Stationary sensor nodes

Component	Number
Rack #01	
Host PC	1
LTE USB-Stick [26]	1

[24] https://www.ardusimple.com/simplertk2b/
[25] https://www.ardusimple.com/product/helical-antenna/
[26] https://consumer.huawei.com/en/mobile-broadband/e3372/

SIM card [27]	1
Cable USB A - A socket ... cm	1
GPSDO SRS FS740 [28]	1
1G ETH Cable (min. 1m) to Host PC	1
GNSS Survey-In Antenna + tripod	1
Kabel (TNC-SMA) (GNSS)	1
SMA Cable 5m	1
USRP X310 with UBX frontends	3
Fiber optics cable (50m) to USRP X310	3
Hyperflex10 N-Type Cable (50m) 1 PPS, 10 MHz	6
N-SMA breakout cable (1 PPS, 10 MHz)	6
Adapter (N-BNC) (1 PPS, 10 MHz)	6
Rx #01, #02:	
High pass [29](2.9-8.7 GHz)	2
LNA (ZX60-83LN-S+[30])	2
incl. 6 V power supply + battery	2
6 dB attenuator (SMA)	2
SMA Cable (60 cm, Mini-Circuits 141-24SM+)	2
SMA Cable (7.5 Meter, Hyperflex10) #01, #02	2
Tx:	
High pass[31](3.9-9.8 GHz)	2
power amplfiier 2-6 GHz (RFHIC RUM43020-10[32])	1
- SN: 1015120003	
lab power supply 12V PA	1
SMA Cable (7.5m, Hyperflex10) #05	1
Antenna (Poynting A-WLAN-0060-V1[33]	3
Antenna mount	3
Tripods (Winch tripods)	3
Antenna booms + antenna	3
Wood pieces for Tripod feet support	9
Screws (e.g. Torx 25)	9
Washer rings	9
Water canisters	9
Ropes	9
Tensioning belts for RXs' safety	2
Cablestraps	
Painter's tile as underlay for canisters	
Rain Cover	
Wastebags for LNA Raincover	2
small semitransparent box as pad for USRP/PA	4
big transparent 60L box as cover for USRP	2
big transparent 120L box as cover for USRP/PA	1

[27] https://geschaeftskunden.telekom.de/startseite/mobilfunk/462274/data-comfort-xs-business.html
[28] https://www.thinksrs.com/downloads/pdfs/catalog/FS740c.pdf
[29] https://www.minicircuits.com/pdfs/VHF-2700A+.pdf
[30] https://www.minicircuits.com/pdfs/ZX60-83LN-S+.pdf
[31] https://www.minicircuits.com/pdfs/VHF-3500+.pdf
[32] http://rfhic.com/eng/prod/view.html?idx=222
[33] https://www.ime.de/shop/www.ime.de/produktdatenblaetter/A40418.pdf

Tarp for rack raincover	1
Power supply 230V	
cable reel 10m	2
cable reel 25m	1
cable reel 50m	1
Tools	
SMA torque wrench	1
fork wrench 8	1
slot screwdriver	1
Torx 25 Screwdriver	1
spirit level	1
measurement stick	1
angular gauge	0.5

Rack #02	
Host PC	1
Power supply 230V cable reel min. ... m	1
10 GB or 100GB Switch	1
LTE USB-Stick [34]	1
SIM card [35]	1
Cable USB A - A socket ... cm	1
GPSDO SRS FS740[36]	1
1G ETH Cable (min. 1m) to Host PC	1
GNSS Survey-In Antenna + magnetic mounting	1
Adapter (BNC-SMA) (GNSS)	1
Adapter (TNC-SMA) (GNSS)	1
SMA Cable 5m	1
USRP X310 with UBX frontends	2
Fiber optics cable (50m) to USRP X310	2
Hyperflex10 N-Type Cable (50m) 1 PPS, 10 MHz	4
N-SMA breakout cable (1 PPS, 10 MHz)	4
Adapter (N-BNC) (1 PPS, 10 MHz)	4
Rx #03, #04:	
High pass [37](2.9-8.7 GHz)	2
LNA (ZX60-83LN-S+[38])	2
incl. 6 V power supply + battery	2
6 dB attenuator (SMA)	2
SMA Cable (60 cm, Mini-Circuits 141-24SM+)	2
SMA Cable (7.5 Meter, Hyperflex10) #03, #04	2
Antenna (Poynting A-WLAN-0060-V1[39]	2
Antenna mount	

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[34] https://consumer.huawei.com/en/mobile-broadband/e3372/
[35] https://geschaeftskunden.telekom.de/startseite/mobilfunk/462274/data-comfort-xs-business.html
[36] https://www.thinksrs.com/downloads/pdfs/catalog/FS740c.pdf
[37] https://www.minicircuits.com/pdfs/VHF-2700A+.pdf
[38] https://www.minicircuits.com/pdfs/ZX60-83LN-S+.pdf
[39] https://www.ime.de/shop/www.ime.de/produktdatenblaetter/A40418.pdf

Table 2.4: Hardware components sensor nodes at EAZ rooftop and parking lot